Supawich Puengdang

supawich@gmail.com • linkedin.com/in/supawich-puengdang-9197781a4/

Data Scientist specializing in deep learning and geospatial analytics. My experience includes leveraging machine learning and AI to build and optimize asset appraisal models, fraud detection systems, and automated AI-powered room decoration. Passionate about exploring the potential of Generative AI to drive innovation within organizations.

Education

DUKE UNIVERSITY, Durham, NC
August 2025 - May 2027
Master of Science, M.S. in Electrical and Computer Engineering (Machine Learning track) (GPA: 3.92)

CHULALONGKORN UNIVERSITY, Bangkok, Thailand
August 2019 - August 2023
Bachelor of Engineering, B.Eng. (Computer Engineering): 2nd Class honors (GPA: 3.58/4.0)

Professional Experience

Professional Data Scientist
KASIKORN LABS COMPANY LIMITED, Nonthaburi, Thailand
June 2023 – July 2025

- Pioneered Deep Learning for Land Appraisal: Directed a team to research and optimize a novel land appraisal model using deep learning and image processing. This innovative approach reduced the appraisal process from two weeks to minutes, cutting costs by up to $500,000 and significantly reducing appraiser workload.
- Automated AI-Powered Room Decoration: Built an automated AI agent system using Flux.1 Kontext and Google Gemini to create a pipeline for text-to-image generation. This system automatically decorated over 100,000 room images on KBank's NPA website, a project estimated to increase upselling and boost sales by at least 40%.

Data Scientist Research Internship
KASIKORN LABS COMPANY LIMITED, Nonthaburi, Thailand
June 2022 – May 2023

- Developed an ensemble model combining CatBoost, XGBoost, and LightGBM algorithms to assess credit risk. Based on questionnaire responses, the model accurately predicted the probability of late payments for loan applicants.

Publications

S. Puengdang, W. Ausawalaithong, P. Nopratanawong, N. Keeratipranon, and C. Wongkamthong, "**Thailand Asset Value Estimation Using Aerial or Satellite Imagery**," TENCON 2023 - 2023 IEEE Region 10 Conference (TENCON), Chiang Mai, Thailand, 2023, pp. 399-404, doi: 10.1109/TENCON58879.2023.10322494.

- Proposed a novel similarity-based asset valuation model utilizing a Siamese-inspired Neural Network with a pre-trained EfficientNet architecture. This method outperformed the baseline model, achieving an AUC of 0.81 and significantly improving recall from 59.26% to 69.55%.

S. Puengdang, S. Tuarob, T. Sattabongkot, and B. Sakboonyarat, "**EEG-Based Person Authentication Method Using Deep Learning with Visual Stimulation**," 2019 11th International Conference on Knowledge and Smart Technology (KST), Phuket, Thailand, 2019, pp. 6-10, doi: 10.1109/KST.2019.8687819.

- Proposed an innovative method for individual authentication by combining steady-state visual evoked potential (SSVEP) and event-related potential (ERP) features using a LSTM network. This method leveraged consumer-grade BCI devices for rapid and efficient small-scale authentication. These are some of my awards from this project:
    - **Winner**: Best of Information Technology Category, Prime Minister's Science Awards 2019
    - **Runner-up**: Young Scientist Competition 2018
    - **First Prize**: ASEAN Student Science Project Competition 2018
    - **Finalist**: Intel International Science and Engineering Fair (Intel ISEF 2018)

Awards

Winner: Highest AUC in the competition
Aihack Thailand 2021 AI x Financial data
December 2021

- Trained an ensemble classification model to predict Long-Overdue Debtor (LOD) from debtor's data.

Winner: Highest Average Precision (AP) (with IoU = 0.5) among 20 Finalist teams
AI and Robotics Ventures Hackathon 2021
December 2021

- Created a synthetic dataset & train a YOLOv5 model to classify underwater objects.

1st Runner-up: The second-lowest Logarithmic Mean Absolute Error (LMAE) of all competitors
Thailand Machine Learning for Chemistry Competition (TMLCC)
October 2021

- Predicted the gas adsorption ability of metal-organic frameworks using multimodal Graph Neural Networks.

Winner
Outstanding National Youth Award in Innovation and Invention
September 2019

- Received an Outstanding Award in Innovation and Invention from the Ministry of Social Development and Human Security.

Skills

- Deep Learning and Machine Learning tools: Scikit-learn, TensorFlow, PyTorch.
- Proficient in data science and analytics tools: Pandas, NumPy, Plotly, BS4, Gradio, etc.
- Experience with geospatial libraries: GeoPandas, Shapely, Folium.
- Software development fundamentals, including data structures, algorithms, and design patterns.
- Programming languages: Python (primary), C, and C++.
- Foundational knowledge of backend web development, including experience with Go, SQL, MongoDB, Docker, and Kubernetes.